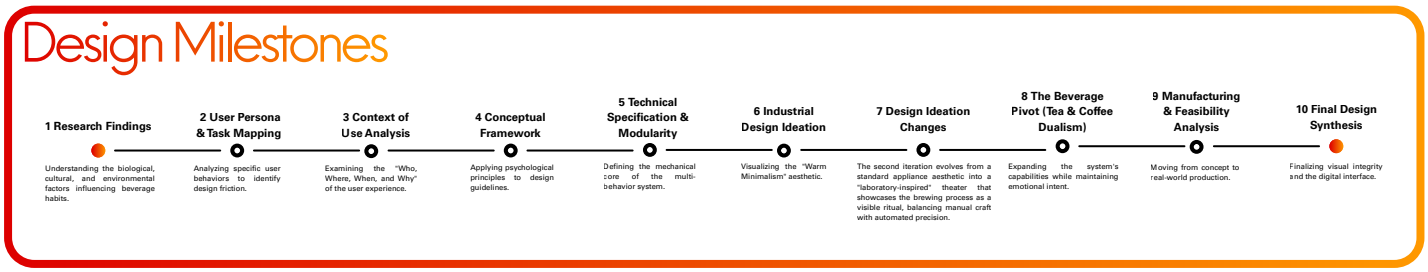


Research Findings

Keywords

- Ritual
- Cognitive Anchor
- IKEA Effect
- Circadian Rhythms.

psychological and biological factors influencing human behavior

Selected impactful prompts

What influences habits related to preparing hot beverages (tea, coffee, herbal teas)?"

Beverage habits are driven by biological triggers (ex:cortisol response) and a "Cognitive Anchor," where brewing serves as a mental transition to work or relaxation.

AI Role

Expander (Further understanding of psychological factors).

Reflections

AI is highly effective in accelerating the initial research phase, but manual source validation is still necessary to avoid errors.

Design Decision

Assuming that ritualistic preparation enhances the perceived value of the beverage, the design goal was set for "meaningful labor."

User Persona & Task Mapping

Keywords

- Task Analysis
- Overflow Anxiety
- Cognitive Load
- Efficiency.

Selected impactful prompts

A "Taste vs. Time" trade-off was identified; users sacrifice taste for multitasking during workdays but prefer traditional methods when time is available.

Comparative Workflow Mapping

Workflow Stage	Teacher (Taste-Driven Ritual)	Herbal Tea Student (Precision Ritual)	Civil Servant (Utility Mode)
Context Trigger	Morning caffeine need while preparing for class	Intentional tea preparation during study or relaxation	Quick hot beverage during work break
Time Availability	Medium → sometimes rushed	High → willing to spend time	Very Low → needs immediate result
Preparation Action	Adds filtered water and coffee to brewing chamber	Selects loose-leaf blend and fills infusion basket	Presses Quick Mode button
Behavior Across the Two Axes	Ritual Mode → visible brewing, manual engagement / Assist Mode → automated but preserves flavor profile	Ritual Mode → visible brewing, manual engagement	Quick Mode → fully automated

AI Role

Descriptive (Categorized "Traditional" and "Automated" workflows).

Reflections

AI helped visualize the emotional and attention-demanding nature of manual brewing (e.g., "overflow anxiety").

Design Decision

The system should be "context-sensitive," acting as a "Craftsman" tool when time is ample and a "Survival" aid when time is limited.

Context of Use Analysis

Keywords

- Persona
- Psychological State
- Environmental Constraints.

Selected impactful prompts

"Make an analyze for the Context of Use... to visualize their emotional state and energy levels.".

User Journey & Emotional State Mapping

User	Core Need	Emotional Driver	Pain Point
56-year-old Teacher	• Strong morning caffeine • Better taste in manual method • Time efficiency on workdays	• “I cannot function without it.” • Belief in authenticity	• Overflow risk when using copper cezve • Machine coffee tastes inferior
25-year-old Student	• Loose-leaf brewing • pH-sensitive brewing • Visual ritual	• Hobbyist passion • Precision & control • Aesthetic satisfaction	• Tea bags are tasteless • Water chemistry changes taste • Industrial look reduces experience
57-year-old Civil Servant	• Hot drink during break • Accessibility • Low maintenance	• Convenience • Functional expectation • Practicality	• Complex settings • Waiting time

AI Role

Re-framer (Shifted focus from simple recipes to psychological dependency).

Reflections

Reframing the user as a "professional in a hurry" highlighted the need for dual-mode operation.

Design Decision

Accepted that "becoming functional" is the primary user goal; success is measured by cognitive readiness for work.

Conceptual Framework

Keywords

- Labor leads to love
- Human-in-the-Loop
- Invisible Guardrail.

Selected impactful prompts

"How can I adapt the given research data to a ritualistic... device [that] should give the user a feeling of putting in effort... but doesn't feel autonomous?".

Context-Adaptive Ritual System

The system doesn’t force users to choose between manual or automatic modes. Instead, it redistributes control through a Gradient Interaction Dial, allowing users to physically express their brewing intent rather than selecting options on a screen.

Dual-Mode Functionality:

- Ritual Mode (time-rich): Manual interaction with a weighted lever and visible brewing to enhance engagement and satisfaction.
- Assist Mode (time-poor): Automated, walk-away safe brewing with sensors and auto-shutoff for reliable, effortless use.

AI Role

Expander (Provided industrial analogies like high-end audio and luxury cars).

Reflections

The concept of "meaningful strain" (ritual) making the reward sweeter was a breakthrough insight.

Design Decision

Rejected fully autonomous buttons in favor of "staged automation" where the user must "approve" key phases of brewing.

Technical Specification & Modularity

Keywords

- PID Induction
- NTC Thermistor
- Borosilicate Glass
- Modularity

Selected impactful prompts

"What is needed for this design concept to work?".

- Unified Technical Architecture
- Heating Core
- Induction-based flash heating
- PID temperature stability
- Vacuum-insulated 2.5L reservoir
- Sensor Layer
- Single-motion cleaning release

- NTC thermistors
- Foam-rise detection
- Context recognition via:
- Interaction speed
- Time-of-day patterns
- Repeated usage behavior
- Modular Brew Chamber
- Removable borosilicate glass
- Stainless infusion module

AI Role

Critique (Identified engineering challenges like cleaning friction and valve sealing).

Reflections

identification of "cleaning friction" as the #1 reason for habit abandonment led to a focus on removable modules.

Design Decision

Accepted the modular brew chamber to preserve user authorship while ensuring easy maintenance.

Industrial Design Ideation

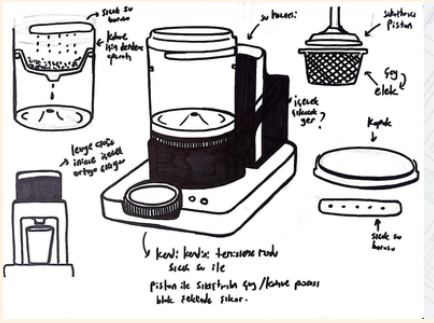
Keywords

- Sculptural Laboratory
- Anti-appliance
- Haptic Texture.

Selected impactful prompts

"Industrial design sketch of a premium context-adaptive hot beverage system... [with] a weighted mechanical rotary dial and a tactile metal lever.". create image

AI visual outputs suggested a minimalist base with a clear borosilicate chamber that acts as a "visible stage" for the brewing ritual. An AI image was requested from the sketch, and then a model was created using Rhino.



AI Role

Expander (Suggested "Warm Minimalism" materials like matte charcoal and ribbed glass).

Reflections

The dial became the primary "Effort Regulator," allowing users to express intent physically rather than through a screen.

Design Decision

-Accepted the weighted mechanical dial as the central interface to provide tactile feedback and agency.

Design Ideation Changes

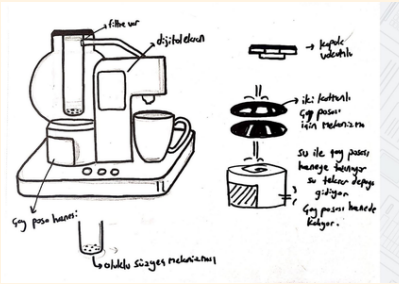
The second iteration shifts from a familiar appliance look to a laboratory-inspired design that makes the brewing process visible and transparent, turning the machine into a brewing theatre that clearly expresses the ritual and the transition between manual and automated control.

Keywords

- Brewing theatre
- Feel like an alchemist
- Anti-appliance

Selected impactful prompts

Create a realistic representation of the given sketch



To better support the ritual–automation balance and the Sculptural Laboratory concept, the design was refined: the Gradient Interaction Lever was replaced with a button, the cylindrical reservoir became a volumetric flask, and the machine was simplified to brew only tea due to technical limitations with combining tea and coffee brewing.

AI Role

Expander (realistic design and material alternatives different from the sketch were provided).

Reflections

Visualizing the developed design helps in observing the technical and physical changes made to the design.

Design Decision

The design reveals the brewing process through borosilicate glass, turning tea making into a visible ritual while supporting natural, temperature-specific preparation for different teas.

The Beverage Pivot (Tea & Coffee Dualism)

Keywords

- Over-steep Bitterness
- Infusion Control
- Visual Bloom

Selected impactful prompts

What features... have changed, and would such a change in concept prevent the machine from working?".

The concept remains sound because both rituals support visible transformation and precision temperature control. The new design was modeled using the Rhino modeling program. rendered with different textures by artificial intelligence.

AI Role

Re-framer (Identified the transition from "Overflow Anxiety" in coffee to "Bitterness Anxiety" in tea).

Reflections

Switching to tea highlighted the importance of a steep-time sensor as a "safety net".

Design Decision

Accepted the addition of a 200-micron steel mesh filter to accommodate various tea leaf sizes and ensure high-quality extraction.

Manufacturing & Feasibility Analysis

Keywords

- CNC Aluminum
- Laboratory Glass Forming
- Food-Grade Silicone

Selected impactful prompts

Internal technical analysis of components

Estimated prototype cost between $130–$200, identifying glass-heating integration and valve sealing as primary engineering hurdles.

1. Mechanical Structure
   - Borosilicate glass brewing chamber
   - Central tea transfer tube
   - Rotary valve system

2. Fluid & Filtration
   - Water pump circulation
   - Micro-perforated filtration
   - Waste and leaf separation

3. Electronics & Control
   - Microcontroller-based system
   - Sensors for temperature and water level
   - Motorized valve control

4. Materials & Manufacturing
   - Food-safe glass, steel, silicone
   - CNC and injection manufacturing

AI Role

Critique (Analyzed costs and manufacturing constraints).

Reflections

Understanding the cost of components like the PID controller and induction coil helped refine the "premium" positioning.

Design Decision

Decided on CNC aluminum and borosilicate glass to meet both aesthetic goals and food safety standards (FDA/LFGB).

Final Design Synthesis

Keywords

- Anti-appliance
- Angled Display
- Translucent Ritual.

Selected impactful prompts

How can I make this tea maker... less mechanical-looking, and allow the glass to see the beverage inside?".

Suggested angling the digital display for better ergonomics and using glass more prominently to emphasize the "alchemy" of the brew.

AI Role

Critique (Evaluated visual balance and interface placement).

Reflections

Removing excess mechanical housing allows the "liquid theatre" to be the focal point of the device.

Design Decision

- Accepted: Induction heating for silent, precise temperature control.
- Rejected: Fully autonomous operation; users must always "trigger" the process.